

CARLOS VELASCO MORENO

MECHATRONICS ENGINEERING · EMERGING TECHNOLOGIES, AI & ROBOTICS

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PROFILE

Mechatronics Engineering student at Tec de Monterrey (GPA 93.67/100) who works hands-on across the emerging-technology stack: artificial intelligence, biometric and motion-capture data, 3D printing and CAD, and embedded electronics. Two-time Shell Eco-marathon medalist with Borregos Racing and a research scholar in the SHIELD group, where I turn raw sensor data into results a supervisor can act on. Fully bilingual and used to demoing and troubleshooting new technology for classmates and faculty. Open to internships and part-time roles in AI, robotics and emerging technologies.

EDUCATION

Tecnológico de Monterrey (ITESM), Campus Monterrey

Aug 2024 – Expected 2028

B.S. Mechatronics Engineering · GPA 93.67 / 100 · 5th Semester

- Coursework: Programming, Mechatronic Integration, AC Circuits, Process Simulation, Differential Equations.

EXPERIENCE

Research Scholar — SHIELD Research Group, Tec de Monterrey

2025 – Present

- Process and analyze biomechanical data from **Vicon motion capture** and **Tekscan pressure-mapping** systems for a study on bowling-throw mechanics.
- Built reusable Python pipelines (Pandas, NumPy, Matplotlib) that clean raw sensor exports, filter out-of-sensor frames and fit 95% confidence ellipses to center-of-force trajectories across nine trials.
- Report and explain findings to the project supervisor; certified in ICH Good Clinical Practice E6(R3).

Vehicle Team Member — Borregos Racing, Tec de Monterrey

Feb 2025 – Dec 2025

- 2nd place at Shell Eco-marathon Americas (Apr 2025) and Brazil (Aug 2025)**, Prototype Battery-Electric, peaking at 373.2 km/kWh — a Mexican energy-efficiency national record in two consecutive competitions.
- Responsible for energy-systems optimization and vehicle subsystem integration, working to international safety and technical regulations under competition deadlines. Built the team's website (HTML/CSS/JS).

PROJECTS

VibeMap — AI Architecture Visualizer

HackaDays, Tec de Monterrey · 2026

- Drop in a project folder and get an interactive Mermaid.js architecture map with per-file drill-down, built so people can understand AI-generated code instead of copy-pasting it. Shipped in an 8-hour hackathon.
- Stack: React, TypeScript, **Gemini API**, Mermaid.js, Tailwind CSS.

QuestBody — Biometric Wellness Wearable & Gamified App

Team project · 2026

- Wristband concept plus a working web prototype in which a simulated 1 Hz **biometric feed** (heart rate, HRV, stress, stillness) triggers RPG quests, XP progression and loot in real time. Team of three; I built the interactive demo and the financial model.

Passive Lower-Limb Exoskeleton — Chairless-Chair Type

Course project · 2026

- Wearable exoskeleton with a pinned knee lock that sends body weight to the floor. I did the concept, the full **SolidWorks** design and FEA (750 N load: 64.59 MPa peak stress vs. 276 MPa yield), and built a **3D-printed** full-scale prototype sized to my own body.

Clinic Operations Simulation — Rockwell Arena

Academic project

- Modeled three simultaneous clinic operations to identify and mitigate bottlenecks, analyzing stochastic arrival and service data to optimize patient flow.

TECHNICAL SKILLS

- Programming:** Python (NumPy, Pandas, Matplotlib, scikit-learn), C/C++, JavaScript / TypeScript, MATLAB
- AI & Data:** Deep learning fundamentals, LLM & API integration (Gemini), AI agent building (Salesforce Agentforce), data analysis and visualization
- Emerging Tech & Hardware:** Arduino and microcontrollers, sensors and electronic instrumentation, 3D printing, SolidWorks (CAD), EasyEDA (PCB design), Vicon motion capture, Tekscan pressure mapping
- Tools & Workshop:** Git / GitHub, Linux (Debian), VS Code, Rockwell Arena, soldering, mill and lathe
- Languages:** Spanish (native) · English (full professional proficiency — able to work and interview entirely in English)

CERTIFICATIONS & AWARDS

- Fundamentals of Deep Learning** — NVIDIA Deep Learning Institute (2025)
- Artificial Intelligence & Agentforce** (20 h) — Iberoamerican Technology Foundation with Salesforce (2026)
- ICH Good Clinical Practice E6(R3)** — The Global Health Network, 100% (2026) · **Science of Exercise** — CU Boulder / Coursera (2026)
- 2x 2nd Place, Shell Eco-marathon** — Prototype Battery-Electric, Americas and Brazil editions (2025)